



5355 - Adapting the Constrained Matrix Factorization Deblender Scarlet to Separate Overlapping Sources in JWST IFU Data

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST's incredible sensitivity is producing exceptionally deep images of the sky. At these depths, the surface density of sources is so high that a large fraction of extended sources overlap with other sources. Separating the light from each source ("deblending") is critical to every science case done with these deep images. This work is challenging with broad-band imaging, but with integral field unit (IFU) data, which is effectively many narrow-band images, we can use spatial and spectral correlations to properly separate all photons into their originating source.

In this context, scarlet, a constrained matrix factorization algorithm originally designed for the Legacy Survey of Space and Time (LSST) pipeline, is an invaluable resource. It excels in disentangling multi-band observations using chromatic and spatial information, surpassing the capabilities of current tools that predominantly rely on single-band deblending.

We propose to adapt and rigorously test scarlet to effectively process spectral data cubes generated by JWST's IFUs, specifically the NIRSpec IFU and MIRI MRS. This adaptation empowers the extraction of individually decomposed spectra from overlapping sources without the necessity of prior templates, thereby advancing the work initiated by the JWST-funded q3dfit project.

While tailored to JWST's IFUs, this program will adapt scarlet for generic IFU usage, thus extending the deblending capacities of the astronomic

community at large. We will develop all tools using NASA's suggested open-source guidelines, providing code to the community as we develop it.

OBSERVING DESCRIPTION